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Clocks from Nuremberg and Augsburg in the 16th and 17th centuries

In early modern Germany, two cities took the lead in the manufacture of clocks and scientific instruments: Augsburg and Nuremberg.

Nuremberg in the 16th century

In the course of the 15th century, the free imperial city of Nuremberg became a center of watch and instrument making. Situated at the crossroads of important trade routes, Nuremberg was one of the largest cities in Germany with extensive trade and flourishing craftsmanship and was a frequent meeting place of the Imperial Diet. Raw materials (copper, zinc, tin, brass and bronze) were easily accessible, and special tools (small files, drills, etc.) could easily be purchased in the city.¹ Nuremberg's metal craftsmen and instrument makers were known for the quality of their work, from which mathematicians, astronomers and cartographers also profited. In 1471 the astronomer Johannes Regiomontanus (1436 – 1476) settled in Nuremberg, established an observatory and a printing house, from which numerous mathematical and

¹ Nuremberg has been designated as the "ancestral city of filecutters" (Dick, Otto: *Die Feile und ihre Entwicklungsgeschichte*, Berlin 1925, pp. 23 – 31).

astronomical works emerged.² His pupil and patron Bernhard Walther (1430 – 1504) was the first to use a clock for astronomical observations.³

In the first half of the 16th century, watchmaking in Nuremberg was a free craft⁴, but some clockmakers joined the locksmiths' guild (*Plattschlosser*). In 1565, three watchmakers (Esaias Vogel, Hanns Praun and Markus Stopinger) were admitted and requirements for the masterpieces were decreed by the city council. The three applicants were obviously experienced craftsmen and had to present a table clock with striking on the hour and quarter hour and indications of calendar data. The other requirement was a small, portable clock with hour strike.⁵ But for all future applicants the table clock was to be equipped additionally with an astrolabe, and the small clock was to possess an alarm mechanism as well as an additional 12 or 24-hour strike. The two timekeepers were to be completed over a period of eight months.⁶

The table clocks manufactured in Nuremberg as well as in Augsburg are described in the literature somewhat vaguely as table-clocks (*Türmchenuhren*) or *Stutzuhren* (a "short", "truncated" clock, relative to longcase clocks). In the second half of the 16th century, a transverse rectangular type with astronomical and calendar indications emerged in southern Germany and the Austrian hereditary territories.⁷ According to the ground plan, these clocks have two wide main fronts and narrow side fronts. The bells of the striking mechanism are housed under an ogival arched roof with apertures and a crowning figure, armillary sphere or the like on its apex.

² Pilz, Kurt: 600 Jahre Astronomie in Nürnberg, Nürnberg 1977, pp. 58ff. – Stromer, Wolfgang v.: Hec opera fient in oppido Nuremberga Germanie ductu Ioannis de Montereio. Regiomontan und Nürnberg 1471 – 1475, in: Günther Hamann (ed.), Regiomontanus-Studien, Wien 1980, pp. 267 – 289 (Sitzungsberichte Österreichische Akademie der Wissenschaften, Phil.-Hist. Kl., 364). – Wingen-Trennhaus, Angelika: Regiomontanus als Frühdrucker in Nürnberg, in: Verein für Geschichte der Stadt Nürnberg: Mitteilungen 78, 1991, pp. 17 – 87.

³ Beaver, Donald deB.: Bernard Walther. Innovator in Astronomical Observation, in: Journal for the History of Astronomy 1, 1970, pp. 39 – 43. – Pilz (n. 2), pp. 93 – 103.

⁴ Hampe, Theodor (ed.): Nürnberger Ratsverlässe über Kunst und Künstler im Zeitalter der Spätgotik und Renaissance, 3 vols., Wien/Leipzig 1904 (Quellenschriften für Kunstgeschichte und Kunsttechnik des Mittelalters und der Neuzeit, n. s., 12 – 13), vol. I, pp. 498 – 499, no. 3518. – Baader, Joseph: Beiträge zur Kunstgeschichte Nürnbergs, in: Jahrbücher für Kunstwissenschaft 1, 1868, p. 259.

⁵ Stockbauer, Jacob: Nürnbergisches Handwerksrecht des XVI. Jahrhunderts. Schilderungen aus dem Nürnberger Gewerbeleben, Nürnberg 1879, pp. 3f. – complete text of the order according to a copy of 1629 in: Gümbel, Albert: Aktenstücke zur Geschichte der mittelfränkischen Uhrmacherkunst, Halle 1928, pp. 1 – 8.

⁶ Stockbauer (n. 5), pp. 3 – 4.

⁷ Neumann, Erwin: Die Tischuhr des Jeremias Metzker von 1564 und ihre nächsten Verwandten, in: Jahrbuch der Kunsthistorischen Sammlungen in Wien 57, 1961, pp. 89 – 122.

Hans (Johannes) Gruber the Elder, born around 1530, is one of Nuremberg's most important watchmakers († 1597).⁸ In 1552 he became a master locksmith (*Plattschlosser*), and in 1583 produced his chief work, a table-clock intended for the Danish King Frederick II.⁹ {Bild001.tif} {Bild002.tif} Actually Gruber was supposed to build a simple travel clock, but he delivered an elaborately designed, magnificent timepiece with an elaborate iconographic program designed specifically for a ruler personality.¹⁰ The gilded brass case was made by a hitherto unknown coppersmith. An even more elaborately designed clock with an astrolabe dial was created in 1584 for Heinrich Rantzau (1526 – 1598), governor of the Danish King Frederick II in the duchies of Schleswig and Holstein. {Bild003.tif} But the clock preserved in Skokloster Castle in Sweden (inv. no. 597)¹¹ is not a bespoke piece for a certain client, because two similarly designed table clocks are extant (Berlin, Kunstgewerbemuseum, inv. no. 99.295¹²; Oslo, Institute for Theoretical Astrophysics). The goldsmiths and engravers commissioned by Gruber (Hans Episcofer and Mathis Zündt) apparently used certain ornamental components to decorate the case. The tympanum of the astrolabe dial also proves that the clock is, so to speak, "made-to-measure": Although it was engraved for the geographic latitude of 54.5° – corresponding to the place of residence of the buyer – there is a network of lines for 50° northern latitude on the back. So the clock was originally intended for the Nuremberg market and was only "converted" after Rantzau had acquired it. The design of the differo-epicyclic gear train¹³ of the astrolabe is identical to that of the astronomical clock on the Käfigtor in Winterthur, built by Laurentius Liechti in 1529.¹⁴ {Bild004.tif} {Bild005.tif} But there is a decisive

⁸ For a list see Maurice, Klaus: *Die deutsche Räderuhr. Zur Kunst und Technik des mechanischen Zeitmessers im deutschen Sprachraum*, 2 vols., München 1976, vol. I, pp. 299 – 300.

⁹ Stuttgart, Landesmuseum Württemberg; Himmelein, Volker/Leopold, John H.: *Prunkuhren des 16. Jahrhunderts. Sammlung Joseph Fremersdorf*, Stuttgart 1974, pp. 59 – 61.

¹⁰ Schaller, Andrea: *Prunkuhren der Renaissance*, Stuttgart 2001, pp. 33 – 39.

¹¹ Oestmann, Günther: *Eine astronomische Uhr aus dem Besitz Heinrich Rantzaus*, in: *Nordelbingen. Beiträge zur Kunst- und Kulturgeschichte Schleswig-Holsteins* 67, 1998, pp. 41 – 55.

¹² Hausmann, Tjark: *Alte Uhren*, Berlin 1979, pp. 113 – 115 (*Kataloge des Kunstgewerbemuseums Berlin*, 8).

¹³ For a historical overview see: Karlsen, Helge Bernhard Johan: *Differensialprinsippet. En historisk oversikt*, in: *Norsk Teknisk Museum*, Volund 1978, pp. 7 – 44.

¹⁴ Schenk, Adolf: *Die astronomische Uhr des Zeitglockenturms in Winterthur*, Winterthur 1941 (78. Neujahrsblatt der Hilfsgesellschaft Winterthur), pp. 23 – 27. – Brunner, Raimond/Schmitt, Herbert: *Die astronomische Uhr von Winterthur. Beschreibung eines interessanten astronomischen Zifferblattes und Zeigerwerkes*, in: *Schriften der „Freunde alter Uhren“ in der Deutschen Gesellschaft für Chronometrie* 35, 1996, pp. 143 – 167.

difference: Gruber swapped the driving wheels for the movement of the rete and the lunar nodes with each other, which results in wrong periods of revolution. However, it is not possible to assemble the wheel set differently, so this is obviously a systematic error.¹⁵

Gruber also made watches (there is a neck watch in the Victoria and Albert Museum, London). The attribution of another clock made in tiny dimensions (Nuremberg, Germanisches Nationalmuseum, inv. no. LGA 3656) is uncertain, however.¹⁶

Augsburg in the 16th century

Augsburg was in no way inferior to Nuremberg when it came to the quality and originality of the watches manufactured there. Here, too, the conditions were excellent for the development of highly developed trades, for thanks to its convenient location, the city had always been an important trans-shipment center, and it was here that the influential families of the Fugger and Welser with their far-reaching economic activities had their headquarters. The Imperial Diets held in Augsburg brought wealthy customers to the city, where Catholics and Protestants had lived together peacefully since the religious peace negotiated in 1555.¹⁷ Augsburg clocks and automaton were also important as diplomatic gifts to the Sublime Porte in Constantinople, the so-called *Türkenverehrung* ("Turkish presents").¹⁸

¹⁵ Thanks to the generous courtesy of Dr. Bengt Kylsberg and Ann Hallström (Skokloster Castle), the author was able to dismantle and examine the clock exhibited at Gottorf Castle in 1997. Very probably the astrolabe of the Gruber clock in Oslo was also equipped with identical gearing. However, the astronomical gear of this clock has three transmission wheels with 37, 24 and 24 teeth, respectively. Since these are made of brass, it is apparently a later modification; see: Karlsen (n. 13), p. 26.

¹⁶ Eser, Thomas: Die älteste Taschenuhr der Welt? Der Henlein-Uhrenstreit, Nürnberg 2014 (Kulturgeschichtliche Spaziergänge im Germanischen Nationalmuseum, 16), p. 181.

¹⁷ Häberlein, Mark: Wirtschaftsgeschichte vom Mittelalter bis zur Gegenwart, in: Augsburger Stadtlexikon, ed. Günther Grünsteudel, 2nd ed. Augsburg 1998, pp. 146 – 161. – Müsch, Irmgard: Augsburg als Marke. Uhren und wissenschaftliche Instrumente aus der Freien Reichsstadt, in: Die Schwaben zwischen Mythos und Marke, Stuttgart 2016, pp. 244 – 252.

¹⁸ Kurz, Otto: European Clocks and Watches in the Near East, London/Leiden 1975 (Studies of the Warburg Institute, 34), pp. 30 – 42. – Mraz, Gottfried: Die Rolle der Uhrwerke in der kaiserlichen Türkenverehrung, in: Die Welt als Uhr. Deutsche Uhren und Automaten 1550 – 1650, ed. Klaus Maurice and Otto Mayr, München/Berlin 1980, pp. 39 – 54.

The history of watchmaking in Augsburg dates back to the second half of the 14th century. In 1441, the first watchmaker was admitted to the Grand Guild of Blacksmiths, where, in addition to iron- and hammersmiths, locksmiths, gunsmiths, copper and copper hammersmiths, cutlers, file cutters, blacksmiths, makers of spurs and stirrups (*Sporer*), armorers and winch makers were incorporated. Two craft statutes of 1552/54 and 1650, which also served as models for other cities (including Nuremberg), defined the position of the clockmakers within the blacksmith's guild, whereby the latter of the two statutes already took into account the separation of clock- and watchmaking.¹⁹

The characteristic design of the Augsburg table clocks is the result of formal and proper specifications. According to the 1577 regulations, the clockmakers had to build a clock as their masterpiece with an astrolabe, the indication of the day's length, a calendar (year and week days) and striking mechanisms²⁰, which explains a certain uniformity in the movement structures as well as the astrolabe dials despite all the different designs of the cases. These specifications were in effect for 155 years (until 1732). The woodcuts and copper engravings of the Nuremberg instrument maker Georg Hartmann (1489 – 1564), whose workshop produced a large number of sundials, globes and astrolabes and who was the first to introduce paper astrolabes²¹, often served as models for the astrolabe dials. {Bild006.tif} {Bild007.tif} {Bild008.tif}

The city could boast – even if only for a short time – having a monumental astronomical clock, which was erected in 1594 in Augsburg Cathedral in a joint effort by the clockmakers Michael and Hans Marquart, the clock and compass maker Tobias Klieber (around 1545 – 1619) and Georg Henisch (1549 – 1618)²², a physician and

¹⁹ Eva Groiss, *Das Augsburger Uhrmacherhandwerk*, in: *Die Welt als Uhr: Deutsche Uhren und Automaten 1550 – 1650*, ed. Klaus Maurice and Otto Mayr, München/Berlin 1980, pp. 64 – 65; 85 – 86 imprint of the two statutes.

²⁰ Groiss (n. 19), pp. 62 – 89, 70 – 71.

²¹ Klemm, Hans Gunther: *Georg Hartmann aus Eggolsheim (1489 – 1564). Ein Mathematiker, Physiker und Instrumentenmacher von europäischem Rang*, Markt Eggolsheim 1989). – *Ibid.*: *Georg Hartmann aus Eggolsheim (1489 – 1564). Leben und Werk eines fränkischen Mathematikers und Ingenieurs*, Forchheim 1990 (*Wissenschaftliche und künstlerische Beiträge des Ehrenbürg-Gymnasiums Forchheim*, 8). – Schmidt, Suzanne Karr Kathleen: *Interactive and Sculptural Printmaking in the Renaissance*, Leiden/Boston 2017 (*Brill's Studies in Intellectual History*, 270/*Brill's Studies on Art, Art History, and Intellectual History*, 21), pp. 205 – 240, 274 – 289.

²² Bobinger, Maximilian: *Alt-Augsburger Kompaßmacher. Sonnen-, Mond- und Sternuhren. Astronomische und mathematische Geräte. Räderuhren*, Augsburg 1966 (*Abhandlungen zur Geschichte der Stadt Augsburg. Schriftenreihe des Stadtarchivs Augsburg*, 16), pp. 90 – 94.

professor at the grammar school near St. Anna.²³ The clock, described in detail in a manuscript²⁴, had the usual division into three parts (calendar/astrolabe dial/automatons). {Bild09.tif} From a travelogue of the early 16th century it emerges that the clock was not quite completed in autumn 1608. The enormous sum of 36,000 guilders had already been spent for its construction, and for the completion further 3000 guilders were necessary.²⁵ Whether it was lacking financial resources or technical problems that caused Bishop Heinrich V of Knöringen to remove the clock from the church building in the following year is not known.

An important customer of the Augsburg clockmaker's workshops was Duke August the Younger of Brunswick and Lüneburg (1579 – 1666), who, besides his passion for books, also had a marked preference for scientific instruments and clocks {Bild010.tif}. In the course of time he assembled a collection of about 40 neck watches and more than 20 larger clocks. The Augsburg art agent Philipp Hainhofer (from 1613 to 1647), later his sons-in-law Johann Martin Hirt and Johann Georg Anckel, acted as intermediaries.²⁶ The extensive, largely preserved correspondence – the correspondence with Hainhofer alone comprises 1,514 letters – represents a valuable and multi-faceted source on the political and cultural life of the 17th century, which also provides insights into the everyday activities and circumstances of the watchmakers commissioned by the Duke.²⁷

Duke August appears as a professionally educated client, but attached less importance to precious materials and representative equipment of his clocks. What was more

²³ Biographical information on Henisch in Spring, Siegfried: Georg Henisch 1549 – 1618. Arzt, Mathematiker, Gräzist, Professor am Gymnasium bei St. Anna in Augsburg, in: Lebensbilder aus dem Bayerischen Schwaben 13, Weißenhorn 1986, pp. 90 – 117.

²⁴ Bayerische Staatsbibliothek München: Cgm 4189; reprinted in Bassermann-Jordan, Ernst v.: Die verlorene Monumentaluhr im Dome zu Augsburg, in: Die Uhrmacherkunst 48, 1929, pp. 179 – 184, 200 – 203, 220 – 225.

²⁵ Walther, Philipp Alexander Ferdinand (ed.): Hans Georg Ernstingers Raisbuch, Tübingen 1877 (Bibliothek des Litterarischen Vereins in Stuttgart, 135), pp. 119f.

²⁶ Milchsack, Gustav: Herzog August d. J. von Braunschweig und sein Agent Philipp Hainhofer 1613 – 1647, in: Braunschweigisches Magazin 26, 1920, pp. 25 – 34. – Otte, Wolf-Dieter: Johann Martin Hirt und die Augsburger Agentur 1647 – 1661, in: Augsburg in der Frühen Neuzeit: Beiträge zu einem Forschungsprogramm, ed. Jochen Brüning and Friedrich Niewöhner, Berlin 1995, pp. 39 – 118 (Colloquia Augustana, 1). – Boström, Hans-Olof: Philipp Hainhofer als Vermittler von Luxusgütern zwischen Augsburg und Wolfenbüttel, *ibid.*, pp. 140 – 157.

²⁷ Herzog August Bibliothek Wolfenbüttel: Cod. Guelf. 83 – 85, 93 – 96, 98 – 100 Nov. 2°; extracts edited by Gobiet, Ronald: Der Briefwechsel zwischen Philipp Hainhofer und Herzog August d. J. von Braunschweig-Lüneburg, München 1984 (Forschungshefte hg. vom Bayerischen Nationalmuseum München, 8).

important to him was good craftsmanship, reliable technology and precise movement.²⁸ Most of the watches he ordered came from Augsburg, but also from other southern German towns such as Kempten or Ulm. About twenty larger clocks are mentioned in the letters as *Stozzer*, *Stozzen-Uhr* or *Tischuhr*. Furthermore, the correspondence with Hainhofer mentions about forty small watches that can be worn on the body. Unfortunately none of these has survived, and from the other, once rich ducal collection of timepieces only three still exist, all of which are kept in the Herzog Anton Ulrich Museum in Brunswick: a drum-shaped clock of Isaak III Habrecht in Strasbourg rolling on a sloping plain, a mirror clock manufactured by Caspar Langenbucher in Augsburg in 1649, and a Year Clock of Hans Buschmann the Elder, also made in Augsburg in 1651/52.²⁹

Buschmann belonged to a prestigious Augsburg dynasty of clockmakers.³⁰ He had worked in Prague as a journeyman of the imperial chamber watchmaker Jost Bürgi (1552 – 1632) and became a master craftsman in Augsburg in 1620. That year also saw the beginning of his work for Duke August. After Buschmann's death in 1662, his sons worked for the Wolfenbüttel court.

Dealing with the older Buschmann proved to be difficult for the Duke's agents, however. It is mentioned that he was rather unmanageable, a drunkard and did not meet agreed delivery deadlines. The last problem was due to the long trial runs of his timekeepers, because Buschmann was anxious to deliver high quality, and for a long time he renounced any auxiliary staff except that of his sons. Both he and his client sometimes overstrained the technology of the time. But despite all problems, Duke August remained forever faithful to his master clockmaker.

In 1652 the Duke asked him what a "Prague clock" might be and Buschmann informed him that he had built one many years ago in Prague. From memory he executed a

²⁸ This is emphasized in several letters to Hainhofer; see Gobiet (n. 27), pp. 153, 195, 235, 269f., 552.

²⁹ Christoph Rohr's rolling ball clock, originally intended for Duke Julius and completed in 1600, was also owned by Duke August and has been preserved (Herzog Anton Ulrich-Museum Braunschweig: Uhr 77).

³⁰ Bobinger, Maximilian: *Kunstuhrmacher in Alt-Augsburg*, Augsburg 1969 (Abhandlungen zur Geschichte der Stadt Augsburg. Schriftenreihe des Stadtarchivs Augsburg, 18), pp. 79 – 97. – Bobinger (n. 22), pp. 133 – 147.

conceptual drawing (*Visierung*), which is preserved {Bild011.tif}.³¹ Said clock had a high rectangular, glazed box of gilded brass with a wooden base. An intermediate winding mechanism (*remontoir*) was housed in the upper part. This device, conceived by Jost Bürgi, served to reduce the influence of gear friction and the varying motive force of the spring on the rate of a clock. The escapement used by Buschmann is the cross beat (*Kreuzschlag*) also invented by Bürgi, which is a variant of the spindle escapement: The two lobes that engaged with the escapement wheel are attached to two adjacent axles connected by two drives, each carrying two weights at its ends (here: cherubim heads) and swinging against each other. The escapement takes its name from the crossing of the arms at each half oscillation. This enabled a more even rate and represented an outstanding invention by Bürgi, which resulted from the necessity of precise time determination in astronomical observations.³²

In Augsburg the design caused quite a stir, and Johann Martin Hirt, Hainhofer's successor as the Duke's art agent, remarked: "As all clockmakers report, such [clocks] are said never to have been made in Augsburg". In 1652 Duke August placed an order for building it, and within a year Buschmann manufactured the clock for 360 guilders. During transport from Augsburg to Wolfenbüttel, however, it was damaged, and the watchmakers commissioned by Duke August, Heberlin and Schneeberger, were unable to make the clock run reliably. In 1656 other repairs were due, but Bushman refused to carry them out because in his opinion others had botched up the clock. Caspar Langenbucher did not dare to go to work on it and returned the clock unrepaired after two years. Afterwards all traces of it were lost.

Parallel to that Buschmann created a "Year Clock" with remontoir for Duke August in 1651/52 {Bild012.tif} {Bild013.tif}.³³ Three further copies were made for Grand Duke Ferdinand II of Tuscany, his brother-in-law Archduke Charles in Innsbruck, and in 1656 for Emperor Ferdinand III, none of which is preserved. The price was 400 guilders each.

³¹ Himmelein, Volker: Die Uhren, in: Sammler Fürst Gelehrter. Herzog August zu Braunschweig und Lüneburg 1579 – 1666, Wolfenbüttel 1979 (Ausstellungskataloge der Herzog-August-Bibliothek, 27), pp. 168 – 169.

³² Bertele, Hans v.: Precision Timekeeping in the pre-Huygens Era, in: Horological Journal 95, 1953, pp. 794 – 816.

³³ Fink, August: Die Uhren Herzog Augusts d. J., Braunschweig 1965 (Kunsthäfte des Herzog-Anton-Ulrich-Museums, 8), pp. 7 – 10. – Himmelein (n. 31), pp. 166f. – Maurice, Klaus/Mayr, Otto (eds.): Die Welt als Uhr. Deutsche Uhren und Automaten 1550 – 1650, München/Berlin 1980, pp. 230 – 232.

The two sons of the elder Buschmann were also involved in the construction of the “Year Clock”. The outer form resembles an Augsburg art cabinet (*Kabinettschrank*) of the 17th century. The gilded brass case is mounted on an ebony base with painted ivory fillings, and the corners are decorated with gilded bronze angelic herms. The windows on the narrow sides allow a glimpse of the very delicate wheelwork of the clock, which has almost inaudible ticking and no striking mechanism. The clock was transported to Wolfenbüttel in 1652 in a box suspended from straps, but arrived damaged and had to be returned. Duke August remarked ironically, that this year clock was very suitable as a day clock. Even after restoration, the clock never worked properly, and running it for a year without interruption proved to be impossible. In 1657 it was handed over for repair to Caspar Langenbucher, who, after apparently futile efforts, returned the clock unrepai red two years later. However, it turned out to be impossible to receive a fee for the time and effort spent. Wolf Griesbeck († probably 1667), appointed by Duke August 1658 as court watchmaker, seems to have finally set the clock in motion.

In 1648 Duke August called Tobias Treffler, a silver and ivory turner from Augsburg, to Wolfenbüttel to teach his sons the art. Through Treffler, his stepson Caspar Langenbucher (*around 1620, †1677/78) came into contact with the Duke. He worked for eight years as a journeyman for the important watchmaker Johann Sayller (1597 – 1668) in Ulm³⁴, became a master in 1648 and supplied various neck watches to the Duke of Brunswick. In 1650 he offered him his “great masterpiece”, a mirror clock {Bild014.tif}. This ornate clock set on one foot resembles a standing mirror and thus received its name, but this design is also called a “monstrance clock.”³⁵

Langenbucher’s clock is of outstanding quality and very well executed. The astrolabe dial is equipped with a sun, moon and dragon hand and has a calendar ring outside. Two superimposed blued and silver-plated hoops indicate the length of day and night. {Bild015.tif} The balance is visible under a crystal glass plate on the back. {Bild016.tif} The spindle cock and bridges of the movement are decorated with delicate à jour

³⁴ Sayller came to Ulm in 1617 and received citizenship in 1624. In 1647 he became guild master and, following Christoph Margraf, Sayller built rolling ball clocks. He utilized the remontoir introduced by Bürgi. Through his art agent Philipp Hainhofer, Duke August the Younger ordered small neck watches from Sayller. On Sayller see the entry in the *Allgemeines Künstlerlexikon* by the author (forthcoming).

³⁵ Fink (n. 33), pp. 11 – 13; Himmelein (n. 31), pp. 165f.; Maurice/Mayr (n. 33), p. 211.

ornaments. Furthermore there are five dials for hour and quarter-hour strikes, months, the days of the week and the length of the day.

However, the case is not signed by Langenbucher's hand. As was usual for a masterpiece, it was made by other specialized craftsmen. According to the extant instructions for the use and adjustment of the clock, no less than a cabinetmaker (for the octagonal base made of ebony), a turner (for the round, metal case parts), a goldsmith (for the repoussé work and chasing), another gold worker (for the enameled dials and the silver rings of the calendar dial), an engraver (for the flower ornaments on the bridges and the alarm clock case), an engraver for lettering, and finally a bookbinder for the (no longer extant) read leather case with gold imprint and metal fittings were employed.

Caspar Langenbucher was considered the best master of his trade alongside Johannes Sayller in Ulm, but is described as a difficult character who had to be handled "like a raw egg".

Developments in the 17th century

The 17th century brought turbulent times and fundamental upheavals. This applies to political developments, everyday life as well as the stormy development of the sciences. The Thirty Years' War began in 1618 and devastated the whole of central Europe.

Augsburg experienced enormous inflation, precious metals could hardly be obtained, and food became scarce. The population had to endure terrible famines, and as if all that was not enough, the plague raged several times. In 1632 the troops of the Swedish king Gustav Adolf were on their way to southern Germany, they occupied Augsburg for three years, and numerous artisans fled the city. War contributions were rigorously collected and eroded the reserves.³⁶ With all these hardships, daily life was often completely disrupted: Many a time Hainhofer complained about the excessive

³⁶ Roeck, Bernd: Als wollt die Welt schier brechen. Eine Stadt im Zeitalter des Dreißigjährigen Krieges, München 1991. – Häberlein, Mark: Vom Augsburger Religionsfrieden bis zum Ende der Reichsfreiheit, in: Augsburger Stadtlexikon, ed. Günther Grünsteudel, 2nd ed. Augsburg 1998, pp. 75 – 96.

consumption of alcohol by the most skilled craftsmen. In spite of everything, there was a boom, because the occupying forces were war profiteers who brought a lot of money into the city, and in these chaotic times the most beautiful, much sought-after crafted objects were created in Augsburg.³⁷

In one of his founding appeals for learned societies, the universal scholar Gottfried Wilhelm Leibniz (1646 – 1716) praised the great importance of the two imperial cities for the development of the mechanical arts: "It could be explicated in length, in which way Augsburg and Nuremberg have been the training place for the art of clockmaking, waterworks, for turners, goldsmiths, compass makers and innumerable works useful and pleasant to human life" (*Was gestalt Augspurg und Nürnberg die Schuhle aller Mechanicorum gewesen, und die Uhren, Waßerkuenste, Dreh- und Gold- und Circel-Schmids arbeit und unzehliche dem Menschlichen leben nuez- und annehmliche wercke in Schwang gebracht, koente nach der Laenge ausgefuehrt werden.*)³⁸

But at the same time Leibniz had to admit signs of decay: "But we are more and more in a state of decline in manufactures, commerce, means, militia, justice and the form of government, and there is no wonder that also sciences and the arts kiss the dust" (*Aber leyder es gehet mit uns in manufacturen, commercien, mitteln, miliz, Justiz, Regierungsform mehr und mehr bergab, da dan kein wunder, daß auch wißenschafften und Kuenste zu boden gehen.*)³⁹

In contrast to the dynamic developments of the new scientific institutions founded in England, France and Italy, the German territories lacked the promotion of art, crafts and science. After the end of the Thirty Years' War, Augsburg's libraries suffered a complete lack of books on physics, natural history and technology.⁴⁰

But it was not only the devastation of war and the lack of fruitful effects from scientific academies that delayed the perception of fundamental innovations. The rigid

³⁷ Fink, August: Das Augsburger Kunsthandwerk und der Dreißigjährige Krieg, in: Augusta 955 – 1955. Forschungen und Studien zur Kultur- und Wirtschaftsgeschichte Augsburgs, ed. Hermann Rinn, s. I. [München] 1955, p. 325.

³⁸ Leibniz, Gottfried Wilhelm: Bedencken von aufrichtung einer Academie oder Societat in Teutschland zu aufnehmen der Kuenste und Wissenschaften, in: *Œuvres de Leibniz publiées pour la première fois d'après les manuscrits originaux*, ed. Louis Alexandre Foucher de Careil, 7 vols., Paris 1861/75, vol. VII, p. 66).

³⁹ Ibid., p. 73.

⁴⁰ Stetten, Paul von d. J.: Kunst-Gewerb- und Handwerks Geschichte der Reichs-Stadt Augsburg, Augsburg 1779/88, vol. I, pp. 74f.

regulations of the guilds also played a role, as can be seen from the introduction of the pendulum.

In 1657 Christiaan Huygens (1629 – 1695) received a privilege for the pendulum clock constructed by him and built by the watchmaker Salomon Coster and immediately published a description in the following year.⁴¹ With the new design, it was possible to increase the accuracy of the clock from about a quarter of an hour per day to about 10 seconds. Duke August soon heard of this revolutionary and momentous invention and immediately wished to own such a clock. In the spring of 1660 he inquired about clocks "which instead of a balance have a pendulum, as they are made in Holland".

In Augsburg the pendulum clock was still completely unknown, and Caspar Langenbucher declared, that he did not want to confuse his head with such inventions. On the other hand, the sons of Buschmann were interested in it, as can be seen from a letter written by Anckel at the end of April 1660. Whether they recognized the significance of the new invention seems questionable, however.

The experiences of Johann Philipp Treffler (1625 – 1698) are also characteristic.⁴² He had learned the craft from his father Tobias, the silver turner already mentioned, but without completing training. He probably owed his knowledge of watchmaking to the Langenbucher family. Around 1650 Treffler went to Florence and worked for Grand Duke Ferdinando II de' Medici as a clockmaker and turner. He took part in experiments of the Florentine Accademia del Cimento and repeatedly supplied instruments. He also studied Galileo's ideas for the construction of a pendulum clock and was able to produce a usable clock according to his design in 1656. In 1664 Treffler returned to Augsburg and in the following year he married Anna-Barbara Kneulin, the daughter of a doctor, thus becoming a member of the *Kaufleutestube*. He remained in contact with the Florentine court and acted as agent of the Medici. However, since he had neither

⁴¹ Huygens, Christiaan: *Horologium*, Den Haag 1658.

⁴² On Treffler see Bedini, Silvio A.: *Johann Philipp Treffler, Clockmaker of Augsburg* s. l., 1957 (offprint of a series of articles in the *Bulletin of the National Association of Watch and Clock Collectors*, 1956/57); *Agent for the Archduke. Another Chapter in the Story of Johann Philipp Treffler Clockmaker of Augsburg*, in: *Physis* 3, 1961, pp. 137 – 158; *The Pulse of Time. Galileo Galilei, the Determination of Longitude, and the Pendulum Clock*, Florenz 1991, pp. 41 – 53; Prinz, Wolfram: *Deutsche Kunstdrechsler am Florentiner Hof. Nachrichten über Johann Philipp und Christoph Treffler, Drechsler und Uhrmacher für Ferdinando II. und Cosimo III. de' Medici, sowie über Jacopo Mariani*, in: *Mitteilungen des Kunsthistorischen Institutes in Florenz* 13, 1967/68, pp. 173 – 184; Keil, Inge: *Johann Philipp Treffler (1625 – 1698), Drechsler und Uhrmacher der Medici*, in: *Zeitschrift des Historischen Vereins für Schwaben* 103, 2011, pp. 157 – 186.

delivered a masterpiece nor been trained as a clockmaker, he repeatedly had disputes with the turners and clockmakers. Treffler claimed that he did not manufacture clocks customary in Augsburg, but rather executed timekeepers using designs based on mathematics. He regarded mathematics as free art and saw himself not subject to any guild regulations. Although Treffler could produce imperial privileges, apparently obtained through the mediation of the Medici, the council of the imperial city caused him all sorts of difficulties by self-confidently insisting on the observance of the guild regulations. His experiences gained in Italy would certainly enable him to bring clockmaking in Augsburg up to date technically.

While the creation of their own craft corporations initially proved useful and beneficial to clockmakers, these now hampered adaptation to technical innovations. The clocks manufactured in Augsburg were distinguished by their outstanding quality, complicated astronomical indications, automaton and extremely magnificent design. In these objects intended for princely or bourgeois cabinets of curiosities, the actual time indication played a subordinate role. However, these clocks generally did not serve the need for greater accuracy arising in the 17th and 18th centuries. The construction drafts were passed on from generation to generation, {Bild017.tif} and in 1702 journeymen complained that they had to make quaint (*altfränkische*) and thus unsalable masterpieces. Thereupon they were notified that they could add “a pendulum and currently fashionable ornaments aimed at a better sale” (*perpendicular und andere zu besserem Verkauf abzweckende Zieraden nach ietziger mode*)⁴³

The southern German territories, which had been leading in the manufacture of clocks and instruments in the 16th century, fell behind, and technical developments in horology shifted more and more to England, France and finally to Switzerland.

Captions

Bild001.tif: Table clock by Hans Gruber, Stuttgart, Landesmuseum Württemberg Stuttgart, inv. Nr. 1973-021

⁴³ Stadtarchiv Augsburg, U III, fol. 391; quoted from Groiss (n. 19), p. 71.

Bild002.tif: Table clock by Hans Gruber, Landesmuseum Württemberg Stuttgart, inv. nr. 1973-021

Bild003.tif: Table clock by Hans Gruber, Nuremberg 1584, Skokloster Castle (Sweden), inv. nr. 597

Bild004.tif: Gear train of the astrolabe gearing of the Gruber table clock (author's drawing)

Bild005.tif: Movement of the astronomical clock on the *Käfigtor* in Winterthur (Schenk, Adolf: The astronomical clock of the Zeitglockenturm in Winterthur, pp. 24/25)

Bild006.tif: Georg Hartmann, Rete of a paper astrolabe, copper engraving, 198 x 145 mm, Herzog August Bibliothek Wolfenbüttel, Ne 284 (2)

Bild007.tif: Weight-driven clock, Augsburg, c. 1560/1570, private property

Bild008.tif: Table clock, Augsburg, c. 1600, private property

Bild009.tif: The astronomical clock in Augsburg Cathedral, assembly of the four watercoloured drawings in Cgm 4189, pp. 2, 4, 7, 32 (Bayerische Staatsbibliothek München)

Bild010.tif: Anonymous (monogramist J. H. K.?), Duke August the Younger (1579 – 1666) in his study, 1656 or 1659, oil on oak wood, 49.3 x 38.8 cm, Herzog August Bibliothek Wolfenbüttel, inv. no. B 10

Bild011.tif: Drawing of a cross-beat clock by Hans Buschmann the Elder, Herzog August Bibliothek Wolfenbüttel, Graph. C 1528

Bild012.tif: Drawing of the "Year Clock" by Hans Buschmann the Elder, Herzog August Bibliothek Wolfenbüttel, Graph. C 1529

Bild013.tif: Hans Buschmann the Elder, "Year Clock", Herzog Anton Ulrich-Museum Braunschweig, Uhr 75

Bild014.tif: Caspar Langenbucher, Mirror Clock (Monstrance Clock), c. 1649, Herzog Anton Ulrich-Museum Braunschweig, Uhr 76

Bild015.tif: Tympanum of the astrolabe of the Mirror Clock by Caspar Langenbucher with Dragon hand

Bild016.tif: Reverse of the Mirror Clock by Caspar Langenbucher

Bild017.tif: Draft by the clockmaker Johann Michael Landtwing (1688 – 1776) for his Augsburg masterpiece, Museum Burg Zug, inv. no. 3296

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